

ADVANCE PROBLEM PROGRAMMING

SLO 1: Define Paradigms

1. Define Procedural, OOP, and Functional Paradigms with Java Examples

Procedural Paradigm: The procedural paradigm organizes a program into functions or procedures. It focuses on a step-by-step sequence of operations.

```
public class Main {  
    static int add(int a, int b) {  
        return a + b;  
    }  
  
    public static void main(String[] args) {  
        System.out.println(add(10, 20));  
    }  
}
```

Output:

30

Object-Oriented Programming (OOP) Paradigm: OOP organizes programs using classes and objects. It combines data and methods that operate on that data.

```
class Calculator {  
    int add(int a, int b) {  
        return a + b;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Calculator c = new Calculator();  
        System.out.println(c.add(10, 20));  
    }  
}
```

Output:

30

Functional Paradigm: The functional paradigm treats computation as the evaluation of functions. Java supports functional programming through lambda expressions and functional interfaces.

```
import java.util.function.BiFunction;  
  
public class Main {  
    public static void main(String[] args) {  
        BiFunction<Integer, Integer, Integer> add = (a, b) -> a + b;  
        System.out.println(add.apply(10, 20));  
    }  
}
```

Output:

30

2. Write the Same Logic Using Both Procedural and Object-Oriented Approaches

Problem: Calculate the area of a rectangle.

Procedural Approach

```
public class Main {  
  
    static int area(int length, int width) {  
        return length * width;  
    }  
  
    public static void main(String[] args) {  
        int length = 10;  
        int width = 5;  
  
        System.out.println("Area = " + area(length, width));  
    }  
}
```

Output:

Area = 50

Object-Oriented Approach

```
class Rectangle {  
    int length;  
    int width;  
  
    Rectangle(int length, int width) {  
        this.length = length;  
        this.width = width;  
    }  
  
    int area() {  
        return length * width;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Rectangle r = new Rectangle(10, 5);  
  
        System.out.println("Area = " + r.area());  
    }  
}
```

Output:

Area = 50

Difference: The procedural approach uses a separate function to perform the calculation, while the object-oriented approach combines the data and related method inside the Rectangle class.

3. Identify Java Features That Support More Than One Paradigm

Java is a multi-paradigm programming language because it supports features from different programming paradigms.

Java Feature | Supported Paradigm | Purpose

Classes and Objects | OOP | Organize data and behavior

Inheritance | OOP | Reuse properties and methods

Interfaces | OOP / Abstraction | Define common behavior

Methods | Procedural / OOP | Perform specific operations

Loops | Imperative / Procedural | Repeat operations

if-else | Imperative / Procedural | Make decisions

Lambda Expressions | Functional | Represent functions

Functional Interfaces | Functional / OOP | Support functional programming

Stream API | Functional | Process collections using functional operations

Recursion | Procedural / Functional | Solve problems through repeated function calls